

Table D-8. Census Bureau Projections Racial Groups

Census Bureau Population Projections	Census Bureau Deaths Projections	Existing Definitions in BenMAP-CE
Definition 1 - White alone - Black alone - American Indian and Alaska Native alone - Asian alone - Native Hawaiian and Other Pacific Islander alone - Two or more races	Definition 1 - White alone - Black alone - American Indian and Alaska Native alone - Asian alone - Native Hawaiian and Other Pacific Islander alone - Two or more races	- White - Black - Native American - Asian
Definition 2 - White alone or in combination - Black alone or in combination - American Indian and Alaska Native alone or in combination - Asian alone or in combination - Native Hawaiian and Other Pacific Islander alone or in combination		

As can be seen in Table D-8, the racial categories available in the Census Bureau projections data do not match those already in BenMAP-CE. Most notably, (1) the *Native Hawaiian and Other Pacific Islander alone* population is not aggregated with *Asian alone* as is done in BenMAP-CE; and (2) *Two or more races* is separated into its own racial group. To map the Census Bureau data into BenMAP-CE definitions, the following steps were taken.

First, the population dataset was used to find proportions to re-assign death and population counts in the *Two or more races* group to the five other racial groups:

1. The *White alone* count was subtracted from the *White alone or in combination* count to find the aggregate number of those that were considered multiracial, who included “White” in their racial identity. This is referred to as the “combination count.” This was completed for each racial group.
2. The “combination counts” for each racial group were added together, to give the total counts by those in the *Two or more races* group. This total is larger than the size of the *Two or more races* group itself, since individuals selected multiple races.

3. Each “combination count” per racial group was divided by the total combination count to create a proportion. This proportion represents the aggregate proportion of combination counts out of all combination counts for a racial group.
4. These proportions were used to assign deaths and population counts in the Two or more races group to the other five groups.

Example:

$(\text{Two or more races total} * \text{Proportion for White group}) + \text{White alone} = \text{White population}$

Finally, the death and population counts for the Native Hawaiian and Other Pacific Islander group was added to the Asian group.

The resulting mortality scaling ratios are shown for select years and age groups in Table D-9.

Table D-9. Ratio of Future All-Cause Mortality Rate to 2017 Estimated All-Cause Mortality Rate

Race	Year	Age Group					
		0-0	5-14	25-34	45-54	65-74	85-99
Asian	2020	0.98	0.99	0.95	0.95	0.98	0.97
	2030	0.92	0.83	0.76	0.82	0.90	0.88
	2040	0.86	0.72	0.63	0.72	0.82	0.83
	2050	0.80	0.63	0.51	0.60	0.76	0.84
	2060	0.75	0.56	0.41	0.51	0.72	0.82
Black	2020	0.95	0.94	0.92	0.91	0.95	0.98
	2030	0.79	0.77	0.66	0.67	0.80	0.89
	2040	0.66	0.63	0.47	0.52	0.66	0.82
	2050	0.56	0.52	0.34	0.40	0.54	0.82
	2060	0.47	0.43	0.24	0.30	0.46	0.83
Native American	2020	0.96	0.96	0.92	0.90	0.94	0.98
	2030	0.83	0.80	0.67	0.69	0.79	0.89
	2040	0.72	0.68	0.49	0.55	0.66	0.83
	2050	0.63	0.58	0.37	0.43	0.56	0.83
	2060	0.56	0.50	0.27	0.34	0.49	0.84
White	2020	0.98	0.97	0.93	0.93	0.97	1.01
	2030	0.92	0.86	0.70	0.73	0.85	0.91
	2040	0.86	0.76	0.54	0.59	0.74	0.86
	2050	0.81	0.68	0.42	0.47	0.64	0.86
	2060	0.76	0.61	0.32	0.38	0.57	0.87

D.1.3.7 Ethnicity-Adjusted, Race-Stratified Mortality Incidence

To estimate all-cause mortality incidence rates stratified by both race and ethnicity, the race-stratified incidence dataset detailed in section D.1.3.5 was adjusted using correction factors developed from regional incidence rates. We downloaded mortality data per Census Region between 2007-2016 from the CDC WONDER mortality database (<https://wonder.cdc.gov/>) for age groups <25 and >25. One set of data was race-stratified incidence for those of Hispanic origin, and the other was race-stratified incidence for all ethnicities. The dataset for race-stratified incidence without Hispanic origin (non-Hispanic) was back-calculated from these datasets.

A correction factor was developed between the Hispanic and non-Hispanic rates for each region, race, and age combination. For example, a Hispanic correction factor for the race group “White” was calculated for a region by dividing the White-Hispanic rate by the White -all-ethnicities rate. A non-Hispanic correction factor was calculated for a region by dividing the White-non-Hispanic rate by the White-all-ethnicities rate. These correction factors were applied to the county-level race-stratified mortality incidence dataset to find both race and ethnicity stratified values.

D.2 Hospitalizations

Hospitalization rates were calculated using data from the Healthcare Cost and Utilization Project (HCUP). HCUP is a family of health care databases developed through a Federal-State-Industry partnership and sponsored by the Agency for Healthcare Research and Quality (AHRQ). HCUP products include the State Inpatient Databases (SID), the State Emergency Department Databases (SEDD), the Nationwide Inpatient Sample (NIS), and the Nationwide Emergency Department Sample (NEDS). HCUP databases can be obtained from the following data services:

HCUP Central Distributor: Many of the HCUP databases are available for purchase through the HCUP Central Distributor. The databases include detailed information for individual discharges, such as primary diagnosis (in ICD-9 codes), patient’s age and residence county. HCUP categorizes hospital admissions in various ways. Hospitalization admissions are reported as emergency (admitted from the emergency department), urgent (admitted from another hospital), elective (admitted from another health facility, including long-term care), newborn (admitted for delivery), trauma (not used by all states), or other/missing/invalid. While a substantial subset of the ISA-identified literature evaluating respiratory hospitalizations restricted analyses to emergency hospital admissions (EHAs), all hospital admission baseline incidence data within BenMAP reflects total hospital admissions due to time constraints limiting the ability to stratify incidence by admission type. In general, the vast majority of respiratory and cardiovascular hospitalizations appear to be emergency or urgent admissions. As such, the total hospital admissions rates in BenMAP should largely align with analogous EHA rates (albeit biased upward due to the small share of hospitalizations that are elective).

